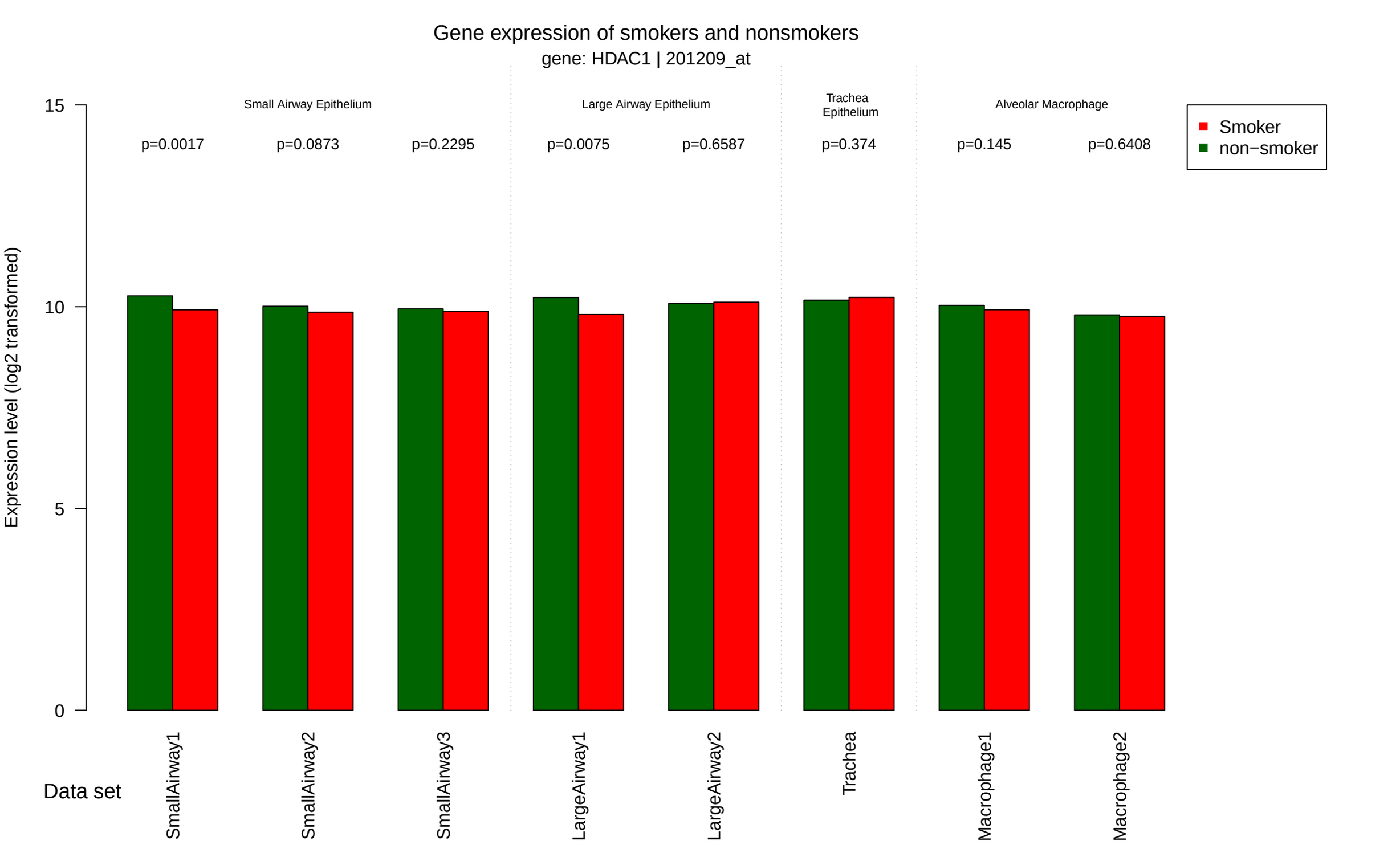
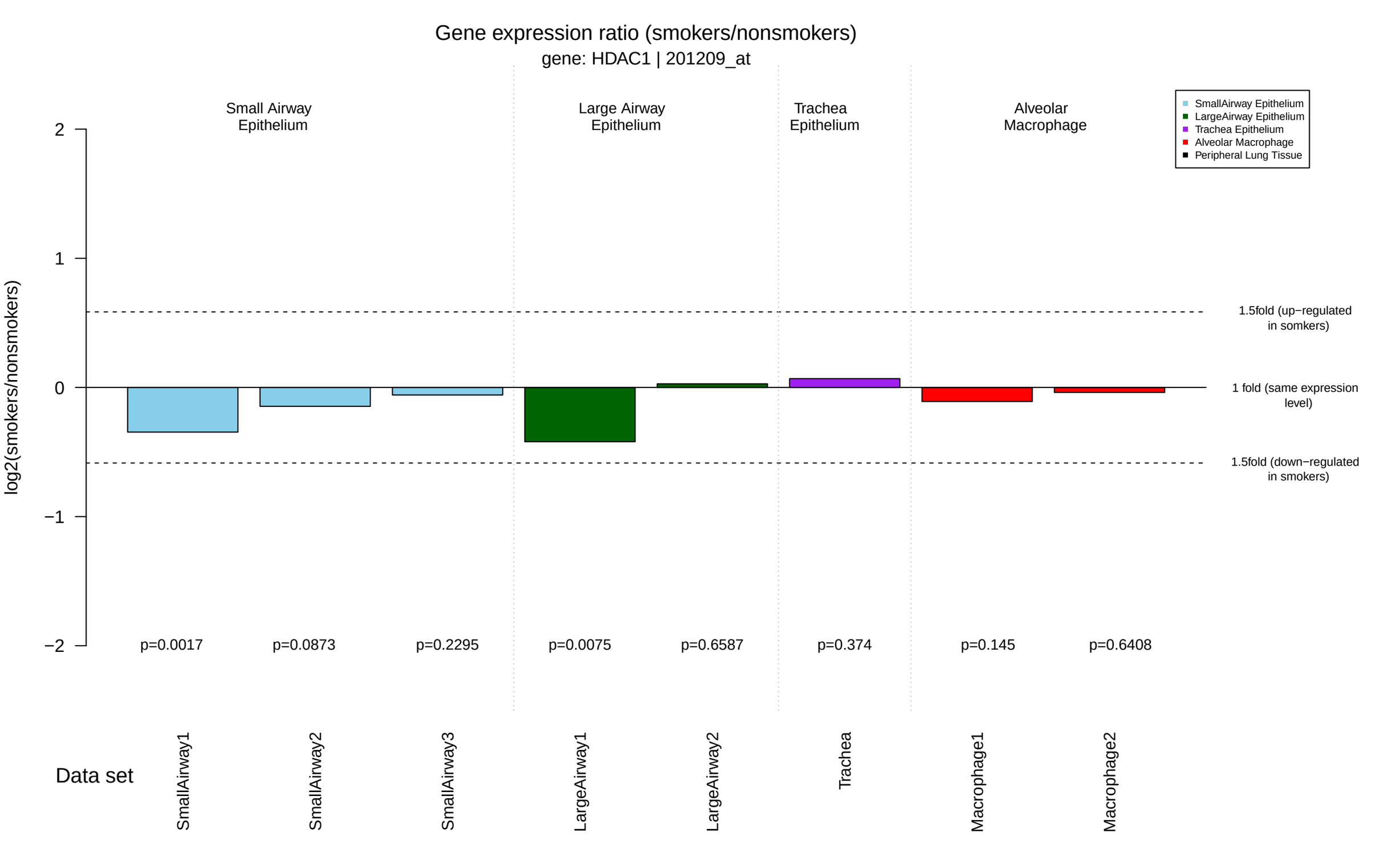
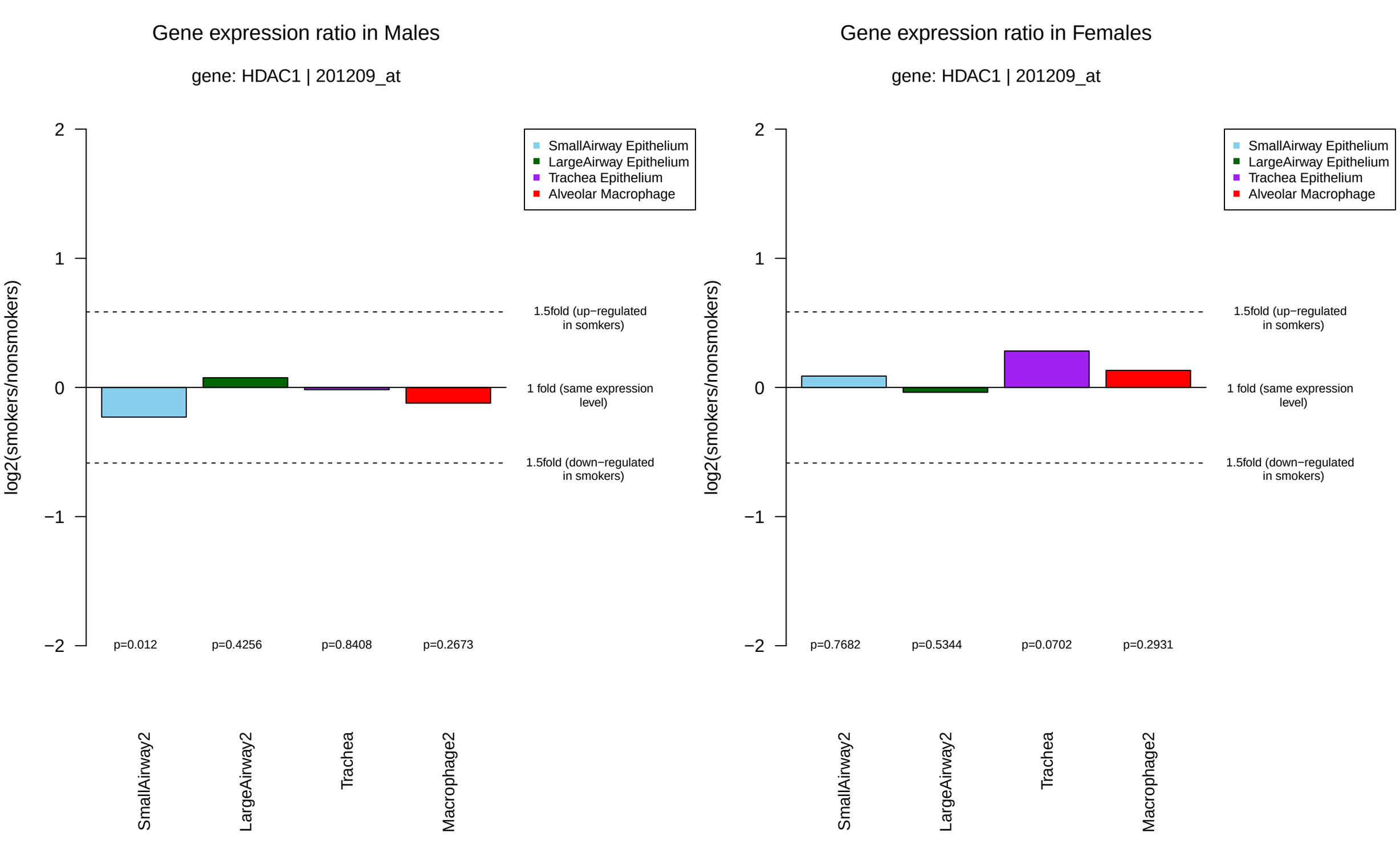
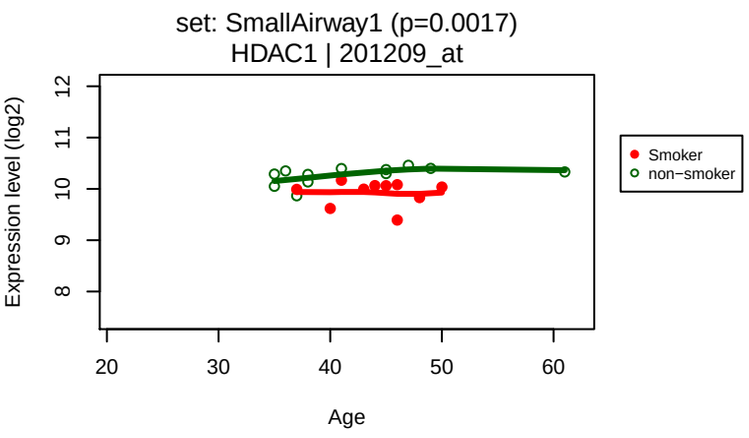
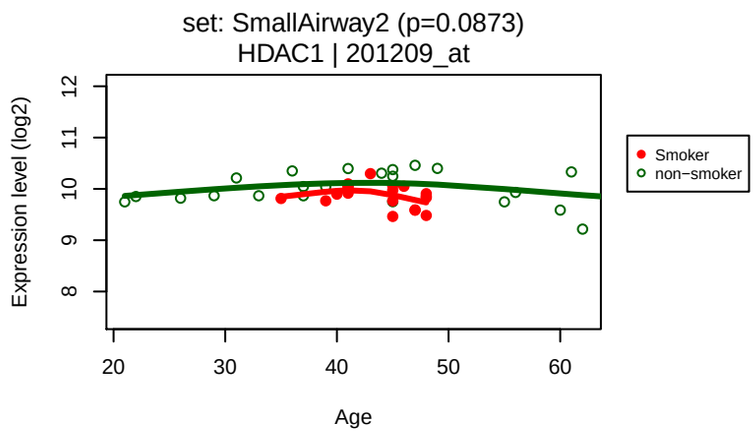
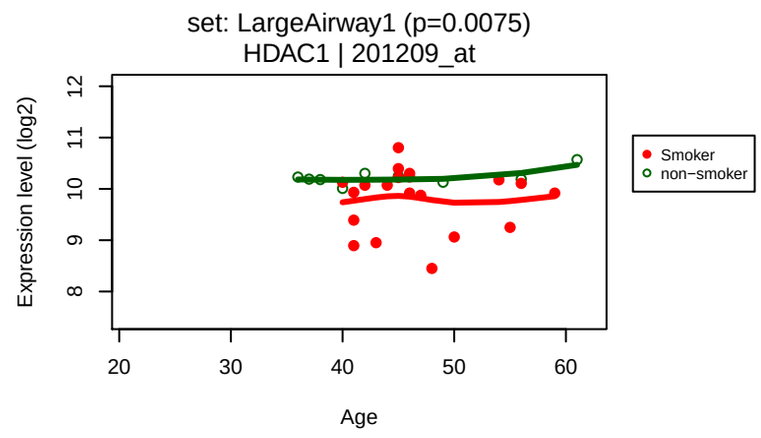

Gene expression of smokers and nonsmokers
gene: HDAC1 | 201209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0017
p=0.0873
p=0.2295
p=0.0075
p=0.6587
p=0.374
p=0.145
p=0.6408
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

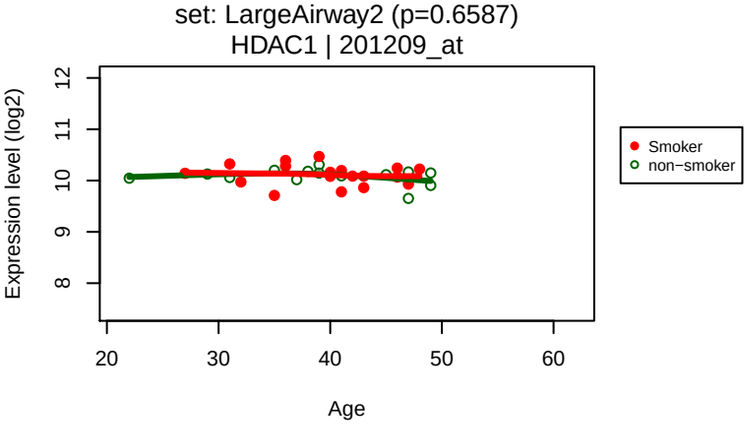

Gene expression ratio (smokers/nonsmokers)
gene: HDAC1 | 201209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0017
p=0.0873
p=0.2295
p=0.0075
p=0.6587
p=0.374
p=0.145
p=0.6408
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

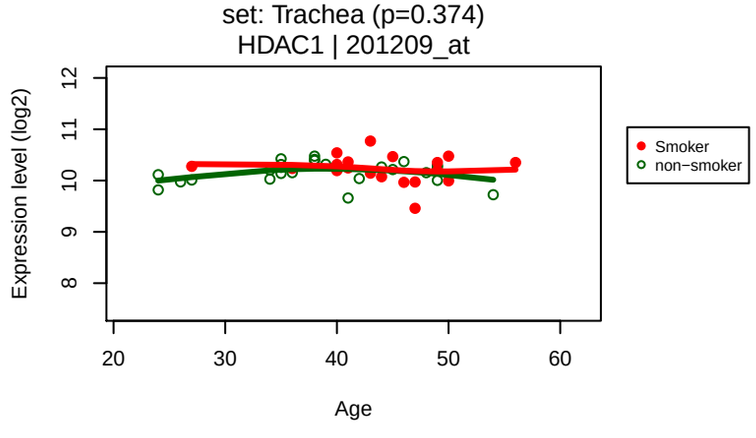

Gene expression ratio in Males
gene: HDAC1 | 201209_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.012
p=0.4256
p=0.8408
p=0.2673
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC1 | 201209_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7682
p=0.5344
p=0.0702
p=0.2931
SmallAirway2
LargeAirway2
Trachea
Macrophage2

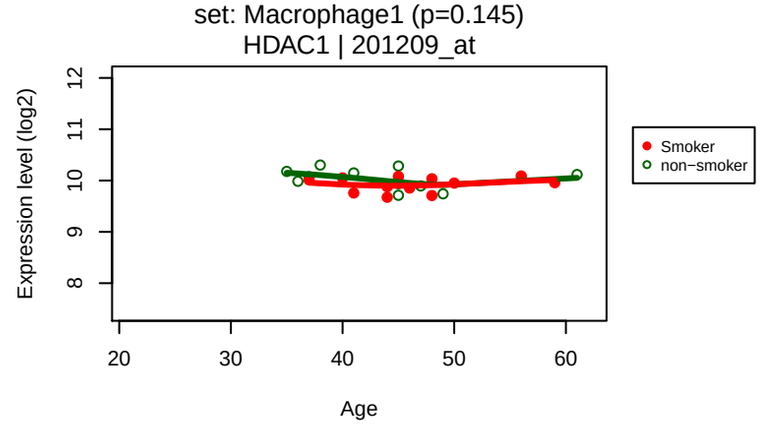

set: SmallAirway1 (p=0.0017)
HDAC1 | 201209_at
Expression level (log2)
Age
Smoker
non−smoker

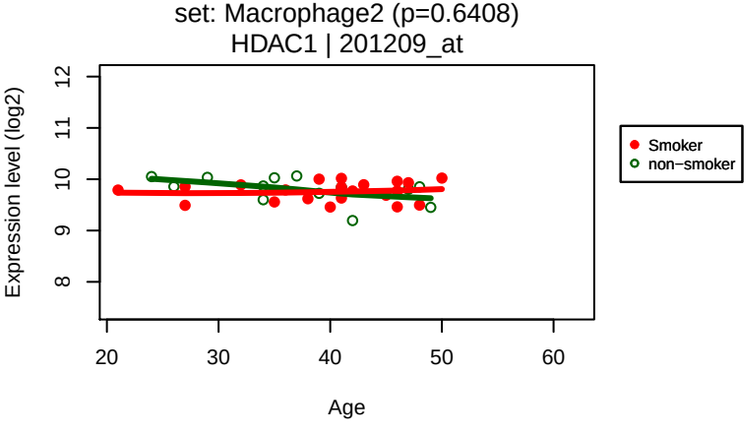

set: SmallAirway2 (p=0.0873)
HDAC1 | 201209_at
Expression level (log2)
Age
Smoker
non−smoker

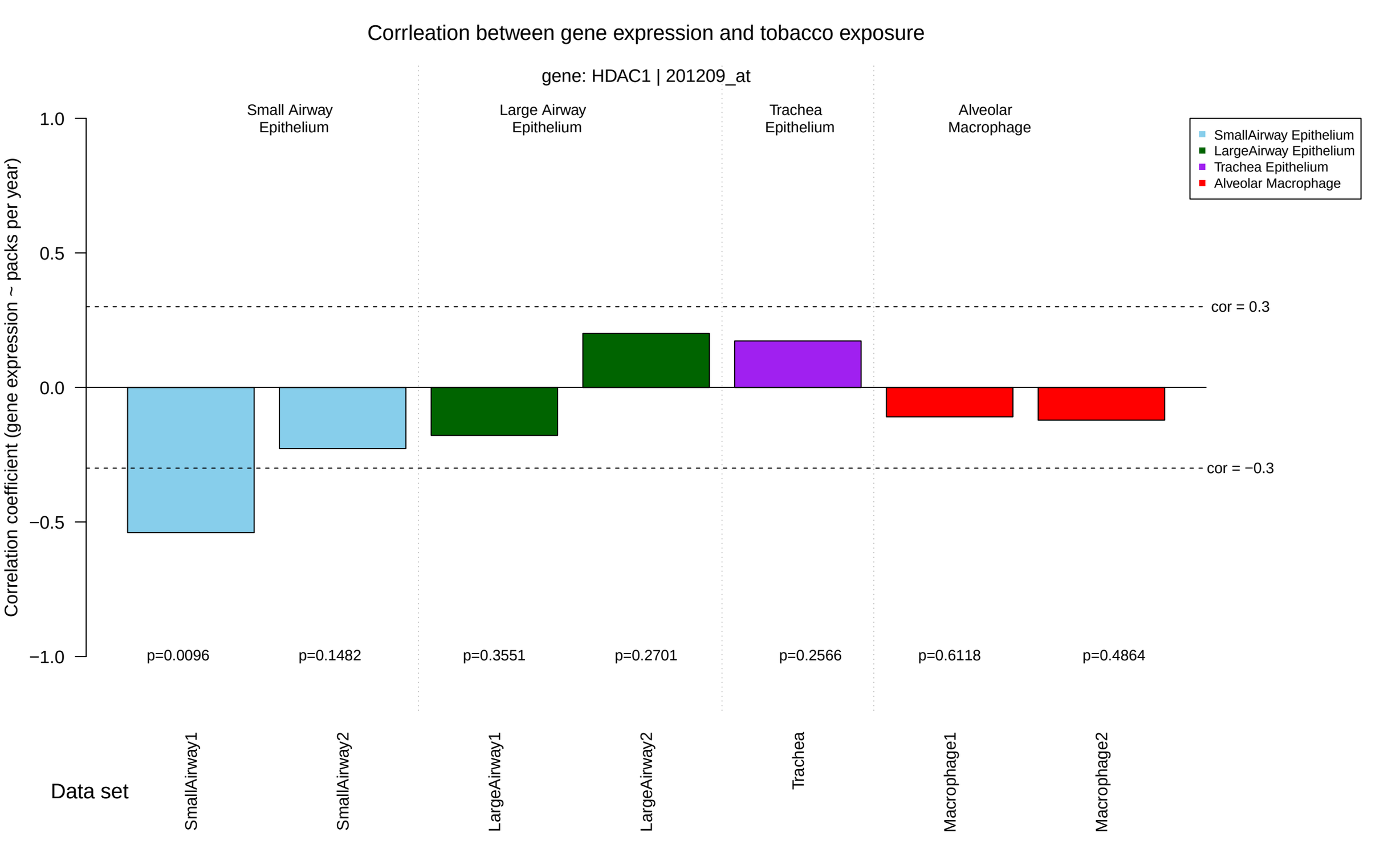

set: LargeAirway1 (p=0.0075)
HDAC1 | 201209_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6587)
HDAC1 | 201209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.374)
HDAC1 | 201209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.145)
HDAC1 | 201209_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6408)
HDAC1 | 201209_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC1 | 201209_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0096
p=0.1482
p=0.3551
p=0.2701
p=0.2566
p=0.6118
p=0.4864
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2